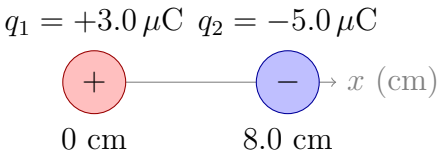
Coulomb's Law Practice

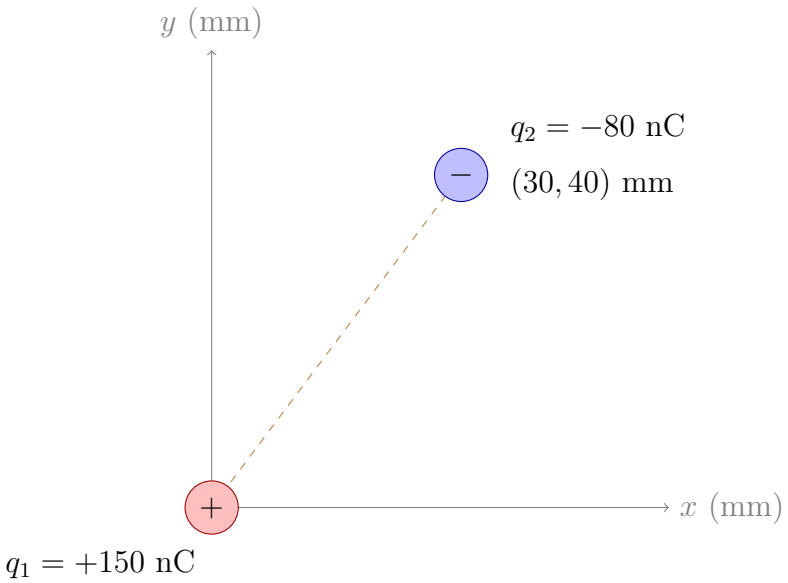
Part A — Separation Vectors & Magnitudes

1. Two charges sit on the x-axis: $q_1 = +3.0\,\mu\text{C}$ at the origin, and $q_2 = -5.0\,\mu\text{C}$ at $x = 8.0$ cm. Find $\vec{r}_{12}$ and $|\vec{r}_{12}|$.



2. $q_1 = +2.0\,\mu\text{C}$ is located at $(1.0\text{ cm}, 2.0\text{ cm})$. $q_2 = -4.0\,\mu\text{C}$ is located at $(5.0\text{ cm}, 5.0\text{ cm})$. Find $\vec{r}_{12}$ and $|\vec{r}_{12}|$.

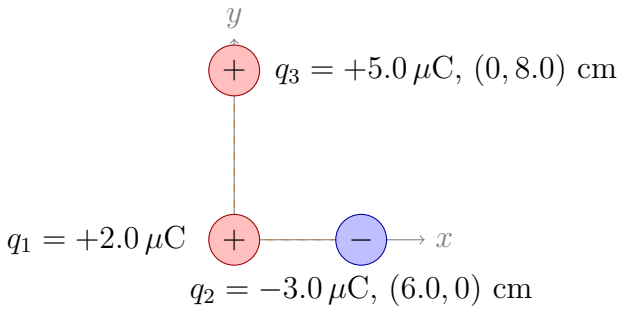
3. $q_1 = +150$ nC sits at the origin. $q_2 = -80$ nC sits at $(30\text{ mm}, 40\text{ mm})$. Find $\vec{r}_{12}$ and $|\vec{r}_{12}|$, in meters.



4. $q_1 = +6.0 \times 10^{-6}$ C is located at $(1.20 \times 10^{-2}$ m, $0)$. $q_2 = -2.5 \times 10^{-6}$ C is located at $(0, 1.60 \times 10^{-2}$ m). Find $\vec{r}_{12}$ and $|\vec{r}_{12}|$.

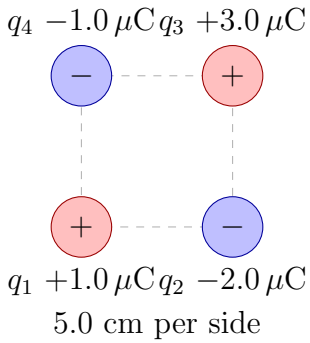
Part B — Separation Vectors & Magnitudes — Multiple Charges

5. Three charges: $q_1 = +2.0\,\mu\text{C}$ at $(0, 0)$, $q_2 = -3.0\,\mu\text{C}$ at $(6.0\text{ cm}, 0)$, $q_3 = +5.0\,\mu\text{C}$ at $(0, 8.0\text{ cm})$. Find $\vec{r}_{12}$ and $\vec{r}_{13}$, and their magnitudes.



6. Three charges: $q_1 = +100$ nC at $(0, 0)$, $q_2 = -50$ nC at $(40\text{ mm}, 0)$, $q_3 = +75$ nC at $(24\text{ mm}, 32\text{ mm})$. Find $\vec{r}_{12}$ and $\vec{r}_{13}$, and their magnitudes, in meters.

7. Four charges sit at the corners of a square, 5.0 cm on a side: $q_1 = +1.0\,\mu\text{C}$ at $(0, 0)$, $q_2 = -2.0\,\mu\text{C}$ at $(5.0, 0)$ cm, $q_3 = +3.0\,\mu\text{C}$ at $(5.0, 5.0)$ cm, $q_4 = -1.0\,\mu\text{C}$ at $(0, 5.0)$ cm. Find $\vec{r}_{12}$, $\vec{r}_{13}$, and $\vec{r}_{14}$, and their magnitudes.



8. Four charges: $q_1 = +4.0 \times 10^{-6}$ C at $(2.0, 1.0) \times 10^{-2}$ m, $q_2 = -3.0 \times 10^{-6}$ C at $(5.0, 5.0) \times 10^{-2}$ m, $q_3 = +2.0 \times 10^{-6}$ C at $(-1.0, 4.0) \times 10^{-2}$ m, $q_4 = -5.0 \times 10^{-6}$ C at $(2.0, -3.0) \times 10^{-2}$ m. Find $\vec{r}_{12}$, $\vec{r}_{13}$, and $\vec{r}_{14}$, and their magnitudes.

Part C — Coulomb Force Magnitude

Same four setups as Part A. Use Coulomb's law, $F = k|q_1 q_2|/r^2$, along with $|\vec{r}_{12}|$, to get the magnitude of the force between the charges.

9. Same as Problem 1: $q_1 = +3.0\,\mu\text{C}$, $q_2 = -5.0\,\mu\text{C}$, separated by 8.0 cm. Find the magnitude of the electric force between them.

10. Same as Problem 2: $q_1 = +2.0\,\mu\text{C}$ at $(1.0, 2.0)$ cm, $q_2 = -4.0\,\mu\text{C}$ at $(5.0, 5.0)$ cm. Find the magnitude of the force between them.

11. Same as Problem 3: $q_1 = +150$ nC, $q_2 = -80$ nC, at the positions given in Problem 3. Find the magnitude of the force between them.

12. Same as Problem 4: $q_1 = +6.0 \times 10^{-6}$ C, $q_2 = -2.5 \times 10^{-6}$ C, at the positions given in Problem 4. Find the magnitude of the force between them.

Part D — Individual Forces

Same four setups as Part B. Find the individual force on q_1 due to each other charge, one pair at a time — magnitude and direction. Don't add them together yet. Careful: the force on q_1 points along $\vec{r}_{21} = \vec{r}_1 - \vec{r}_2 = -\vec{r}_{12}$, not $\vec{r}_{12}$ — you found $\vec{r}_{12}$ in Part B, but the force on q_1 needs the vector that points at q_1*, the other way.*

13. Same as Problem 5: $q_1 = +2.0\,\mu$C at $(0, 0)$, $q_2 = -3.0\,\mu$C at $(6.0, 0)$ cm, $q_3 = +5.0\,\mu$C at $(0, 8.0)$ cm. Find the force on q_1 due to q_2 alone, and the force on q_1 due to q_3 alone.

14. Same as Problem 6: $q_1 = +100$ nC at $(0, 0)$, $q_2 = -50$ nC at $(40, 0)$ mm, $q_3 = +75$ nC at $(24, 32)$ mm. Find the force on q_1 due to q_2 alone, and the force on q_1 due to q_3 alone.

15. Same as Problem 7 (the square): $q_1 = +1.0\,\mu\text{C}$, $q_2 = -2.0\,\mu\text{C}$, $q_3 = +3.0\,\mu\text{C}$, $q_4 = -1.0\,\mu\text{C}$, 5.0 cm apart along each side. Find the force on q_1 due to q_2 alone, due to q_3 alone, and due to q_4 alone.

16. Same as Problem 8: $q_1 = +4.0 \times 10^{-6}$ C, $q_2 = -3.0 \times 10^{-6}$ C, $q_3 = +2.0 \times 10^{-6}$ C, $q_4 = -5.0 \times 10^{-6}$ C, at the positions given in Problem 8. Find the force on q_1 due to q_2 alone, due to q_3 alone, and due to q_4 alone.

Part E — Net Force

Same four setups one last time. Add the individual forces you found in Part D as vectors to get the net force on q_1. Report both the components and the magnitude.

17. Using your answers from Problem 13, find the net force on q_1 (magnitude and direction).

18. Using your answers from Problem 14, find the net force on q_1 (magnitude and direction).

19. Using your answers from Problem 15, find the net force on q_1 (magnitude and direction).

20. Using your answers from Problem 16, find the net force on q_1 (magnitude and direction). This is the hardest one on the sheet — nice work getting here.

ANSWER KEY — FLIP THE PAGE OVER (OR STAND ON YOUR HEAD)

Part A — Separation Vectors
1. $\vec{r}_{12} = (8.0, 0)$ cm; $|\vec{r}_{12}| = 8.0$ cm $= 0.080$ m
2. $\vec{r}_{12} = (4.0, 3.0)$ cm; $|\vec{r}_{12}| = 5.0$ cm $= 0.050$ m
3. $\vec{r}_{12} = (30, 40)$ mm; $|\vec{r}_{12}| = 50$ mm $= 0.050$ m
4. $\vec{r}_{12} = (-1.20, 1.60) \times 10^{-2}$ m; $|\vec{r}_{12}| = 2.00 \times 10^{-2}$ m

Part B — Separation Vectors, Multiple Charges
5. $\vec{r}_{12} = (6.0, 0)$ cm, $|\vec{r}_{12}| = 6.0$ cm $\cdot$ $\vec{r}_{13} = (0, 8.0)$ cm, $|\vec{r}_{13}| = 8.0$ cm
6. $\vec{r}_{12} = (40, 0)$ mm, $|\vec{r}_{12}| = 40$ mm $\cdot$ $\vec{r}_{13} = (24, 32)$ mm, $|\vec{r}_{13}| = 40$ mm
7. $\vec{r}_{12} = (5.0, 0)$cm, $|\cdot| = 5.0$cm $\cdot$ $\vec{r}_{13} = (5.0, 5.0)$cm, $|\cdot| = 7.07$cm $\cdot$ $\vec{r}_{14} = (0, 5.0)$cm, $|\cdot| = 5.0$cm
8. $\vec{r}_{12} = (3.00, 4.00) \times 10^{-2}$m, $|\cdot| = 5.00 \times 10^{-2}$m $\cdot$ $\vec{r}_{13} = (-3.00, 3.00) \times 10^{-2}$m, $|\cdot| = 4.24 \times 10^{-2}$m $\cdot$ $\vec{r}_{14} = (0, -4.00) \times 10^{-2}$m, $|\cdot| = 4.00 \times 10^{-2}$m

Part C — Force Magnitude
9. $F = 21.1$ N (attractive)
10. $F = 28.8$ N (attractive)
11. $F = 0.0432$ N $= 43.2$ mN (attractive)
12. $F = 337$ N (attractive)

Part D — Individual Forces on q_1
13. from q_2: $(15.0, 0)$ N, 15.0 N $\cdot$ from q_3: $(0, -14.0)$ N, 14.0 N
14. from q_2: $(0.0281, 0)$ N, 28.1 mN $\cdot$ from q_3: $(-0.0253, -0.0337)$ N, 42.1 mN
15. from q_2: $(7.19, 0)$ N $\cdot$ from q_3: $(-3.81, -3.81)$ N, 5.39 N $\cdot$ from q_4: $(0, 3.60)$ N
16. from q_2: $(25.9, 34.5)$ N, 43.2 N $\cdot$ from q_3: $(28.3, -28.3)$ N, 40.0 N $\cdot$ from q_4: $(0, -112)$ N

Part E — Net Force on q_1
17. $\vec{F}_{net} = (15.0, -14.0)$ N; $|\vec{F}_{net}| = 20.5$ N, 43.2° below $+x$
18. $\vec{F}_{net} = (0.00281, -0.0337)$ N; $|\vec{F}_{net}| = 33.8$ mN, 85.2° below $+x$
19. $\vec{F}_{net} = (3.38, -0.218)$ N; $|\vec{F}_{net}| = 3.38$ N, 3.7° below $+x$
20. $\vec{F}_{net} = (54.1, -106)$ N; $|\vec{F}_{net}| = 119$ N, 63.0° below $+x$